IHD — Pretest Probability & Triage

1. Low PTP → no test / CTA

WHAT YOU SEE

Yellow lane sign 'LOW PTP → no stress test or CTA'

WHAT IT ENCODES

For stable chest pain, pretest probability (PTP) of obstructive CAD is estimated from age, sex, and symptom type (typical vs atypical vs non-anginal) per Diamond-Forrester / updated ESC tables. In LOW PTP (e.g. <5–15%), the high negative predictive value of coronary CT angiography makes CTA the best rule-OUT tool, or often no testing at all because disease is so unlikely.

BOARD TRAP

WRONG answer: ordering an exercise/stress test in a very-low-PTP patient. Discriminator — by Bayes' theorem, a positive result in a low-prevalence population is far more likely a FALSE positive than true disease, leading to unnecessary catheterization. Low PTP calls for CTA (rule-out) or no test, not provocative functional testing.

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2. Intermediate PTP → functional

WHAT YOU SEE

'INTERMEDIATE PTP → stress imaging or CTA'

WHAT IT ENCODES

INTERMEDIATE PTP (roughly 15–65–85%) is the sweet spot where non-invasive testing changes the post-test probability most. Use functional stress imaging (stress echo, SPECT/PET MPI, or stress cardiac MRI) to detect inducible ischemia, or coronary CTA to assess anatomy. Choose imaging-based stress (not plain exercise ECG) when the baseline ECG is uninterpretable (LBBB, paced, LVH with strain, digoxin, pre-excitation).

BOARD TRAP

WRONG answer: relying on a plain exercise treadmill ECG when the resting ECG is uninterpretable (LBBB/paced/WPW/LVH-strain) or the patient cannot exercise. Discriminator — those baselines make ST changes unreadable, so you need an IMAGING modality, and non-ambulatory patients need a PHARMACOLOGIC stress (regadenoson/adenosine/dipyridamole for perfusion, dobutamine for echo).

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3. High PTP → angiography

WHAT YOU SEE

'HIGH PTP → straight to angiography'

WHAT IT ENCODES

With HIGH PTP (e.g. >85%, or limiting typical angina with clear high-risk features), proceed directly to invasive coronary angiography — non-invasive testing adds little because a negative result cannot reliably exclude disease at high prevalence, and angiography offers diagnosis plus the option of revascularization in one step.

BOARD TRAP

WRONG answer: insisting on a stress test before cath in a high-PTP patient with limiting typical angina. Discriminator — at high prevalence a negative non-invasive test is most likely a FALSE negative and only delays definitive care; the post-test probability barely moves. Go straight to angiography (and define anatomy/revascularization).

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4. <2 hours: emergency

WHAT YOU SEE

Top-left clock '<2 HOURS' over the red lane

WHAT IT ENCODES

The triage clock separates ACS acuity tiers. The RED <2-hour lane is the very-high-risk, emergent pathway: hemodynamic instability, refractory ischemia, life-threatening arrhythmia, acute HF, or arrest — these require immediate invasive management, distinct from the elective PTP-based outpatient workup of STABLE chest pain.

BOARD TRAP

WRONG answer: applying the stable-chest-pain PTP/stress-testing algorithm to a patient in the <2h emergent lane. Discriminator — pretest-probability triage is for STABLE, undifferentiated chest pain; an unstable/very-high-risk ACS patient bypasses risk scoring and goes straight to the cath lab.

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5. Cardiogenic shock / arrest

WHAT YOU SEE

Red lane boat 'CARDIOGENIC SHOCK / ARREST', warning triangles

WHAT IT ENCODES

Hemodynamic instability (cardiogenic shock, SBP <90 with hypoperfusion) or cardiac arrest is a very-high-risk feature mandating IMMEDIATE invasive coronary angiography (<2h) plus resuscitation, vasopressors/inotropes, and consideration of mechanical circulatory support (IABP, Impella). This is emergent management, never outpatient risk stratification.

BOARD TRAP

WRONG answer: sending an unstable/shocked patient for an elective stress test or CTA. Discriminator — provocative testing is contraindicated and dangerous in instability; these patients need emergent revascularization. Stress testing is for STABLE patients only.

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6. <24 hours pathway

WHAT YOU SEE

Orange clock '<24 HOURS'

WHAT IT ENCODES

The AMBER/<24-hour lane is the high-risk-but-not-arresting NSTE-ACS pathway: confirmed NSTEMI (positive troponin), dynamic ECG changes, or GRACE >140 warrant an EARLY invasive strategy with angiography within 24 hours, but without the minute-to-minute urgency of the red <2h tier.

BOARD TRAP

WRONG answer: deferring a high-risk NSTEMI (GRACE >140, dynamic troponin) to an outpatient/elective workup. Discriminator — high-risk NSTE-ACS belongs in the <24h early-invasive lane, not the green selective lane; conversely it does NOT need the <2h emergent pathway unless instability is present.

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7. Troponin + dynamic ECG

WHAT YOU SEE

Orange boat 'TROPONIN + DYNAMIC ECG'

WHAT IT ENCODES

A rising/falling troponin together with dynamic ECG changes (new ST depression or T-wave inversion) defines higher-risk NSTE-ACS and escalates the patient into the early-invasive (<24h) tier. The DYNAMIC change — not a single absolute value — is what confirms acute myocardial injury and drives escalation.

BOARD TRAP

WRONG answer: discharging or down-triaging a patient whose troponin is rising and ECG is evolving. Discriminator — dynamic biomarker plus ECG change is a high-risk signature requiring admission and early angiography; a static single value or a normal resting ECG alone does not provide reassurance here.

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8. Repeat troponin in 6h

WHAT YOU SEE

Runner carrying clock, 'REPEAT TROPONIN IN 6H' with red X

WHAT IT ENCODES

Serial high-sensitivity troponins (0 and 1h, 0 and 2h, or 0 and 3h depending on assay protocol) are used to rule in or rule out MI when the first value is non-diagnostic; the absolute delta between draws confirms or excludes acute injury. Modern hs-cTn algorithms make the old fixed 6-hour repeat unnecessary in most pathways.

BOARD TRAP

WRONG answer: defaulting to a single repeat troponin at 6 hours when a validated rapid 0/1h or 0/2h hs-troponin algorithm is available. Discriminator — rapid serial protocols rule in/out faster and more accurately; the red X in the scene flags the rigid 6h-only approach as outdated. (Also wrong: ruling out MI off one negative early draw without any serial value.)

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9. Selective green lane

WHAT YOU SEE

Green 'SELECTIVE' sign over the green lane

WHAT IT ENCODES

The GREEN selective lane is the lowest-acuity, ischemia-guided pathway for STABLE patients with negative serial troponins, no dynamic ECG changes, and low risk scores. These patients undergo selective/elective evaluation — non-invasive stress testing or CTA — rather than urgent invasive angiography.

BOARD TRAP

WRONG answer: placing a troponin-positive NSTEMI or unstable patient in the green selective lane. Discriminator — the selective/ischemia-guided strategy is ONLY for the genuinely low-risk, ruled-out patient; any positive troponin or instability mandates the amber (<24h) or red (<2h) lane instead.

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10. Criteria already met

WHAT YOU SEE

'CRITERIA ALREADY MET' speech bubble

WHAT IT ENCODES

When diagnostic criteria for ACS/MI are ALREADY satisfied (e.g. unequivocal dynamic troponin rise with ischemic symptoms/ECG), additional confirmatory non-invasive testing is unnecessary and only delays treatment — proceed directly to the appropriate invasive pathway.

BOARD TRAP

WRONG answer: ordering a stress test or CTA to 'confirm' an already-diagnosed NSTEMI before treating. Discriminator — once MI criteria are met, further diagnostic testing wastes time and adds risk; the next step is antithrombotic therapy plus risk-stratified angiography, not more diagnostics.

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11. Send home

WHAT YOU SEE

Green lane sign '1 — SEND HOME' with red X

WHAT IT ENCODES

Discharge home is appropriate ONLY for truly low-risk patients who have been formally ruled out (negative serial hs-troponins, non-ischemic ECG, low HEART/GRACE score) — and even then with timely outpatient follow-up and any indicated functional/anatomic testing arranged.

BOARD TRAP

WRONG answer: discharging a chest-pain patient before completing serial troponins and risk stratification, especially if the first ECG was normal. Discriminator — premature discharge of an under-worked-up ACS is the classic 'missed MI' error; a single normal ECG or single negative troponin does NOT clear a patient for home.

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12. CTA in active ACS

WHAT YOU SEE

'2 — CORONARY CTA IN ACTIVE ACS' with red X

WHAT IT ENCODES

Coronary CT angiography is best suited to STABLE, low-to-intermediate-risk patients as a rule-OUT tool given its excellent negative predictive value; it is the WRONG tool in active, high-risk ACS, where dynamic disease, the need for revascularization, and the superior diagnostic/therapeutic capability of invasive angiography make CTA inadequate.

BOARD TRAP

WRONG answer: choosing CTA to evaluate an actively ischemic, troponin-positive, or unstable patient. Discriminator — CTA shows anatomy but cannot revascularize, is degraded by tachycardia/arrhythmia common in ACS, and delays definitive care; high-risk active ACS needs INVASIVE angiography, not CTA.

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13. Wait for more labs

WHAT YOU SEE

'3 — WAIT FOR MORE LABS' with red X

WHAT IT ENCODES

Passive 'wait and see' is the wrong response for a higher-risk or unstable patient — management should be driven by the triage tier (red <2h, amber <24h, green selective), not by indefinitely awaiting more data when the clinical picture already warrants action.

BOARD TRAP

WRONG answer: delaying treatment to 'wait for more labs' in a patient who already meets criteria for an escalated tier. Discriminator — once instability or a positive dynamic troponin is present, the next step is to ACT (escalate, give antithrombotics, arrange angiography); waiting passively delays time-critical care and is a tested distractor.
